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Manifest Reference

Every Noctalia plugin must include a `manifest.json` file that describes the plugin and its capabilities.

Complete Example

```
{  
  "id": "my-plugin",
```

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```
  "version": "1.2.3",  
  "minNoctaliaVersion": "3.6.0",  
  "author": "Your Name",  
  "license": "MIT",  
  "repository": "https://github.com/noctalia-dev/noctalia-plugins",  
  "description": "A comprehensive example plugin demonstrating all features",  
  "entryPoints": {  
    "main": "Main.qml",  
    "barWidget": "BarWidget.qml",  
    "desktopWidget": "DesktopWidget.qml",  
    "desktopWidgetSettings": "DesktopWidgetSettings.qml",  
    "controlCenterWidget": "ControlCenterWidget.qml",  
    "launcherProvider": "LauncherProvider.qml",  
    "panel": "Panel.qml",  
    "settings": "Settings.qml"  
  },  
  "dependencies": {  
    "plugins": ["other-plugin"]  
  },  
  "metadata": {  
    "commandPrefix": "mycommand",  
    "defaultSettings": {  
      "enabled": true,  
      "customValue": "hello"  
    }  
  }  
}
```

Required Fields

id

- **Type:** String
- **Required:** Yes

- **Description:** Unique identifier for your plugin
- **Rules:**

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- Use lowercase with hyphens (kebab-case)
- Should be globally unique
- Cannot be changed after publication

Example: "my-plugin", "privacy-indicator", "launcher-button"

name

- **Type:** String
- **Required:** Yes
- **Description:** Human-readable plugin name displayed in the UI

Example: "Privacy Indicator", "Hello World"

version

- **Type:** String
- **Required:** Yes
- **Format:** Semantic versioning (x.y.z)
- **Description:** Plugin version number
- **Rules:**
 - Must follow semantic versioning
 - Format: MAJOR.MINOR.PATCH
 - Used for update detection

Example: "1.0.0", "2.3.1"

author

- **Type:** String
- **Required:** Yes

- **Description:** Plugin author name or organization

Example: "Noctalia Team", "John Doe"

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description

- **Type:** String
- **Required:** Yes
- **Description:** Brief description of what the plugin does
- **Recommendation:** Keep under 100 characters

Example: "A privacy indicator widget that shows when microphone, camera, or screen sharing is active."

Optional Fields

minNoctaliaVersion

- **Type:** String
- **Default:** No minimum
- **Format:** Semantic versioning (x.y.z)
- **Description:** Minimum Noctalia Shell version required

Example: "3.6.0"



Always specify minNoctaliaVersion to ensure compatibility with your plugin's required APIs.

license

- **Type:** String
- **Default:** None
- **Description:** SPDX license identifier

- **Common values:** "MIT", "GPL-3.0", "Apache-2.0"

Example: "MIT"

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repository

- **Type:** String (URL)
- **Default:** None
- **Description:** Git repository URL for your plugin

Example: "https://github.com/noctalia-dev/noctalia-plugins"

Entry Points

The `entryPoints` object specifies which components your plugin provides. All entry points are optional, but you must provide at least one.

```
{
  "entryPoints": {
    "main": "Main.qml",
    "barWidget": "BarWidget.qml",
    "desktopWidget": "DesktopWidget.qml",
    "controlCenterWidget": "ControlCenterWidget.qml",
    "launcherProvider": "LauncherProvider.qml",
    "panel": "Panel.qml",
    "settings": "Settings.qml"
  }
}
```

main

- **File:** Main.qml
- **Purpose:** Background logic and IPC handlers
- **Use cases:**
 - Process IPC commands

- Run background tasks
- Monitor system events

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Example:

```
{
  "entryPoints": {
    "main": "Main.qml"
  }
}
```

See [IPC Guide](#) for details.

barWidget

- **File:** BarWidget.qml
- **Purpose:** Widget displayed in the top/bottom bar
- **Use cases:**
 - Display system information
 - Show indicators
 - Provide quick actions
 - Open panels on click

Example:

```
{
  "entryPoints": {
    "barWidget": "BarWidget.qml"
  }
}
```

See [Bar Widget Guide](#) for details.

desktopWidget

- **File:** DesktopWidget.qml
- **Purpose:** Draggable widget on the desktop background

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- Display at-a-glance information
- Clock, weather, or media displays
- Audio visualizers
- Decorative elements

Example:

```
{  
  "entryPoints": {  
    "desktopWidget": "DesktopWidget.qml"  
  }  
}
```

See [Desktop Widget Guide](#) for details.

desktopWidgetSettings

- **File:** DesktopWidgetSettings.qml
- **Purpose:** Per-instance settings UI for desktop widgets
- **Use cases:**
 - Configure individual widgets
 - Per-instance options that differ from plugin-wide settings

Example:

```
{  
  "entryPoints": {  
    "desktopWidgetSettings": "DesktopWidgetSettings.qml"  
  }  
}
```

**Tip**

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available, when the user opens the widget settings from the context menu.

See [Desktop Widget Guide](#) for details.

controlCenterWidget

- **File:** ControlCenterWidget.qml
- **Purpose:** Button widget in the Control Center panel
- **Use cases:**
 - Quick toggle actions
 - Open plugin panels
 - System shortcuts

Example:

```
{
  "entryPoints": {
    "controlCenterWidget": "ControlCenterWidget.qml"
  }
}
```

See [Control Center Widget Guide](#) for details.

launcherProvider

- **File:** LauncherProvider.qml
- **Purpose:** Search/browse provider for the launcher
- **Use cases:**
 - Custom search sources
 - Command handlers (e.g., >kaomoji)
 - Category-based browsing

Example:

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```
entryPoints : {  
  "launcherProvider": "LauncherProvider.qml"  
},  
"metadata": {  
  "commandPrefix": "mycommand"  
}  
}
```

 Tip

When providing a launcher provider, set `commandPrefix` in `metadata` to define the `>prefix` command. This is also used by `pluginApi.toggleLauncher()` for IPC integration. See [IPC Guide](#) for details.

See [Launcher Provider Guide](#) for details.

panel

- **File:** Panel.qml
- **Purpose:** Full-screen overlay panel
- **Use cases:**
 - Detailed information display
 - Complex UI interactions
 - Configuration interfaces
 - Content browsers

Example:

```
{  
  "entryPoints": {  
    "panel": "Panel.qml"  
  }  
}
```

See [Panel Guide](#) for details.

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- **File:** Settings.qml
- **Purpose:** Settings UI integrated into Noctalia Settings
- **Use cases:**
 - User configuration
 - Plugin customization
 - Feature toggles
 - Preferences

Example:

```
{  
  "entryPoints": {  
    "settings": "Settings.qml"  
  }  
}
```

Coming Soon

Settings Guide - Learn how to create configuration interfaces for your plugin.

Dependencies

Specify plugin dependencies in the dependencies object.

```
{  
  "dependencies": {  
    "plugins": ["other-plugin", "required-plugin"]  
  }  
}
```

plugins

- **Type:** Array of plugin IDs
- **Default:** []

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Example:

```
{
  "dependencies": {
    "plugins": []
  }
}
```

Note

Plugin dependency resolution is not yet implemented. This field is reserved for future use.

Metadata

The metadata object contains additional plugin information.

```
{
  "metadata": {
    "commandPrefix": "mycommand",
    "defaultSettings": {
      "message": "Hello World",
      "backgroundColor": "#A9AEFE",
      "enabled": true,
      "updateInterval": 1000
    }
  }
}
```

commandPrefix

- **Type:** String

- **Default:** Plugin ID
- **Description:** The command prefix used to activate this plugin's launcher provider

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- Defines the >prefix command that activates your launcher provider
- Used by `pluginApi.toggleLauncher()` to pre-fill the search text
- Falls back to the plugin ID if not specified

Usage:

```
{  
  "metadata": {  
    "commandPrefix": "kaomoji"  
  }  
}
```

With the above, the launcher command would be >kaomoji and `pluginApi.toggleLauncher()` would open the launcher with >kaomoji pre-filled.

Tip

Set `commandPrefix` when your desired command is shorter or different from your plugin ID. For example, the calibre-provider plugin uses `"commandPrefix": "cb"` so users type >cb instead of >calibre-provider.

defaultSettings

- **Type:** Object
- **Default:** {}
- **Description:** Default values for plugin settings
- **Purpose:**
 - Provide initial values when plugin is first enabled
 - Serve as fallback when settings file doesn't exist
 - Document available settings

Usage in plugin:

```
// Get value from settings with fallback to default  
readonly property string message:
```

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```
pluginApi?.manifest?.metadata?.defaultSettings?.message ||  
""
```

Supported types:

- String: "hello"
- Number: 42, 3.14
- Boolean: true, false
- Object: {"key": "value"}
- Array: ["item1", "item2"]

Validation Rules

The PluginRegistry validates manifests with these rules:

1. **Required fields:** id, name, version, author, description, entryPoints
2. **Version format:** Must match x.y.z pattern (e.g., 1.0.0)
3. **Entry points:** At least one entry point must be specified
4. **ID format:** Should be lowercase with hyphens

Best Practices

Versioning

- Start at 1.0.0 for initial release
- Increment PATCH for bug fixes: 1.0.1
- Increment MINOR for new features: 1.1.0
- Increment MAJOR for breaking changes: 2.0.0

Description

- Be concise (under 100 characters)
- Focus on what the plugin does

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- Avoid marketing language

Default Settings

- Provide sensible defaults
- Document all available settings
- Use descriptive key names
- Include comments in your code explaining settings

Repository

- Link to the plugin's source code
- Include a README.md in your plugin directory
- Maintain a changelog
- Provide screenshots if applicable

Example Manifests

Minimal Bar Widget

```
{  
  "id": "simple-widget",
```

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```
  "version": "1.0.0",  
  "author": "Your Name",  
  "description": "A minimal bar widget",  
  "entryPoints": {  
    "barWidget": "BarWidget.qml"  
  },  
  "dependencies": {  
    "plugins": []  
  },  
  "metadata": {  
    "defaultSettings": {}  
  }  
}
```

Full-Featured Plugin

```
{  
  "id": "hello-world",
```

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```
  "version": "1.0.0",  
  "minNoctaliaVersion": "3.6.0",  
  "author": "Noctalia Team",  
  "license": "MIT",  
  "repository": "https://github.com/noctalia-dev/noctalia-plugins",  
  "description": "A simple Hello World plugin demonstrating functionalities.",  
  "entryPoints": {  
    "main": "Main.qml",  
    "barWidget": "BarWidget.qml",  
    "desktopWidget": "DesktopWidget.qml",  
    "desktopWidgetSettings": "DesktopWidgetSettings.qml",  
    "controlCenterWidget": "ControlCenterWidget.qml",  
    "panel": "Panel.qml",  
    "settings": "Settings.qml"  
  },  
  "dependencies": {  
    "plugins": []  
  },  
  "metadata": {  
    "defaultSettings": {  
      "message": "Hello World",  
      "backgroundColor": "#A9AEFE"  
    }  
  }  
}
```

See Also

- [Getting Started](#) - Create your first plugin
- [Bar Widget Guide](#) - Develop bar widgets
- [Desktop Widget Guide](#) - Create desktop widgets
- [Desktop Widget Settings Guide](#) - Create desktop widgets settings
- [Control Center Widget Guide](#) - Add control center buttons

- [Launcher Provider Guide](#) - Extend the launcher
- [Panel Guide](#) - Create overlay panels

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